Data Cleaning and Preparation: Student PPT Content

Slide-by-slide content for the student deck, with a hands-on activity after every module. Text under each slide title is what appears on the slide. Lines starting with *Visual* are design suggestions only.

Opening

Slide 1: Title

- Data Cleaning and Preparation
- Excel, Google Sheets, Power Query and AI
- Trainer name | Smart Yugam Academy

Slide 2: Why This Matters

- Analysts spend most of their time preparing data, not analysing it
- Dirty data gives wrong charts and wrong decisions
- Garbage in, garbage out
- Today: clean it, combine it, automate it

Slide 3: What You Will Learn Today

1. Data quality issues
2. Cleaning in Excel and Sheets
3. Sorting, filtering and lookups
4. Power Query
5. AI-assisted data preparation
6. Capstone challenge

Slide 4: Setup Checklist

- Excel 2021 or Microsoft 365 (Google Sheets as backup)
- New workbook with two sheets: **Orders_Raw** and **Products**

- Practice dataset typed exactly as shown, mistakes included
- Rule of the day: never edit raw data; always clean a copy

Slide 5: Practice Dataset (Orders_Raw)

OrderID	Customer	City	OrderDate	ProductCode	Qty	Price	Email
1001	ravi kumar	chennai	05/03/2026	P01	2	450	ravi.k@mail.com
1002	PRIYA SHARMA	Coimbatore	2026-03-07	P02		1200	priya@mail
1003	Anil Verma	MADURAI	7 March 2026	P01	3	450	anil.v@mail.com
1002	PRIYA SHARMA	Coimbatore	2026-03-07	P02		1200	priya@mail
1004	meena iyer	Chennai	09-03-2026	P03	1	N/A	meena.i@mail.com
1005	Suresh Babu	Trichy	10/03/2026	P02	-2	1200	suresh.b@mail.com
1006	ravi kumar	Chennai	12/03/2026	P09	5	300	ravi.k@mail.com
1007	Divya R	coimbatore	13/03/2026	P01	4	Rs.450	divya@mail.com

Slide 6: Practice Dataset (Products)

ProductCode	ProductName	Category
P01	Arduino Kit	Electronics
P02	Raspberry Pi	Electronics
P03	Sensor Pack	Components
P04	Jumper Wires	Components

Module 1: Data Quality Issues

Slide 7: Why Data Goes Bad

- Many people type into many systems
- Forms allow free text

- Old exports and merged files bring mismatches
- Cleaning is a normal step in every project

Slide 8: The Five Problem Types

1. **Duplicates:** same record more than once
2. **Missing values:** blanks or placeholders like N/A
3. **Inconsistent formats:** dates, capitalisation, spellings
4. **Errors and outliers:** impossible or suspicious values
5. **Structural problems:** numbers stored as text, extra spaces, many values in one cell

Slide 9: Examples From Our Dataset

Problem	Example
Duplicate	Order 1002 appears twice
Missing	Blank Qty, N/A price
Inconsistent	chennai / Chennai, five date styles
Error	Qty of -2, email priya@mail, code P09
Structural	"Rs.450" in the Price column

Slide 10: Quick Visual Tests

- Status bar: if Count differs from Numerical Count, some numbers are text
- Left-aligned numbers or dates usually mean text
- Filter dropdown shows spelling variants
- Conditional Formatting, Highlight Duplicate Values

Slide 11: The Four Cleaning Habits

1. Keep the raw data untouched
2. Decide the rule first (Proper Case, yyyy-mm-dd)
3. Record every change
4. Check after cleaning: rows before vs after

Slide 12: HANDS-ON 1: Spot the Dirt (15 min)

1. Copy Orders_Raw: right-click the tab, Move or Copy, tick Create a copy, name it **Orders_Clean**
2. In pairs, add column I called **Issue Notes**
3. Write the problem next to every affected row
4. Count total problems found; the highest correct count wins
5. Hint: look at spaces, case, dates, numbers, blanks, duplicates, codes

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Module 2: Cleaning in Excel and Google Sheets

Slide 13: Text Functions

Function	Use
TRIM	Remove extra spaces
CLEAN	Remove hidden characters
PROPER, UPPER, LOWER	Fix capitalisation
LEN	Count characters
SUBSTITUTE	Replace text inside a cell
LEFT, MID, RIGHT, FIND	Extract part of text
VALUE	Convert text to number

Slide 14: Formulas in Action

- `=TRIM(B2)` turns " ravi kumar " into "ravi kumar"
- `=PROPER(TRIM(CLEAN(B2)))` gives "Ravi Kumar"
- `=SUBSTITUTE(G9,"Rs.","")` removes the text
- `=VALUE(SUBSTITUTE(G9,"Rs.",""))` makes it a number
- `=LEFT(H2,FIND("@",H2)-1)` gets the email username
- Read nested formulas from the inside out

Slide 15: Google Sheets Extras

- Same functions work: TRIM, PROPER, SUBSTITUTE, LEFT, FIND
- `=SPLIT(H2,"@")` splits text into cells
- `=ARRAYFORMULA(PROPER(TRIM(B2:B)))` applies to the full column
- `=UNIQUE(A2:H)` removes duplicates without touching the original

Slide 16: HANDS-ON 2A: Text Cleaning (10 min)

On Orders_Clean:

1. Create **Customer_Clean** and **City_Clean** using PROPER and TRIM
2. Create **Email_Clean** using LOWER and TRIM
3. Create **Price_Clean** to turn "Rs.450" into 450
4. Create **Email_User** to extract the text before the @
5. Use LEN to prove Anil Verma now has one space

Slide 17: Formulas to Values

- Cleaned columns are formulas that depend on the originals
- Copy, then Paste Special, Values
- Delete the old columns only after pasting values

Slide 18: Fixing Dates

- Data, Text to Columns, choose Date and DMY, Finish
- `=DATEVALUE(D4)` for text like "7 March 2026"
- Format as `yyyy-mm-dd` (Format Cells, Custom)
- Test: change format to General; a real date shows a number
- Is 05/03/2026 the fifth of March or the third of May? Always confirm the source format

Slide 19: Duplicates and Blanks

- Flag first, delete later: `=COUNTIFS($A$2:A2,A2,$B$2:B2,B2)`
- Data, Remove Duplicates (tick the columns that define a duplicate)
- Go To Special, Blanks selects empty cells
- `=COUNTBLANK(F2:F9)` counts blanks
- `=IF(G5="N/A","",G5)` replaces placeholders

- Ask the owner: is a blank zero or unknown?

Slide 20: Data Validation

- **List:** dropdown of allowed cities
- **Whole number:** Qty must be 1 or more
- **Date:** between 1 Jan 2026 and today
- **Custom:** `=COUNTIF($A:$A,A2)=1` blocks duplicate IDs
- Circle Invalid Data highlights existing bad values
- Best cleaning is the cleaning you never need to do

Slide 21: Find and Replace

- Ctrl+H
- Wildcards: `*` any text, `?` one character, `~` literal symbol
- Match entire cell contents to avoid damaging P10 while replacing P1
- Use Find Next before Replace All

Slide 22: HANDS-ON 2B: Clean the Orders (15 min)

1. Remove the duplicate order with the flag method; rows fall from 8 to 7
2. Convert all dates to true dates in yyyy-mm-dd
3. Highlight the blank and negative Qty; write your rule in Issue Notes
4. Add validation to Qty and City_Clean
5. Save as **Orders_Clean_v1.xlsx**

Check: Does your value look right, or is it right? (Is 450 a number or text?)

Module 3: Sorting, Filtering and Lookups

Slide 23: Sorting

- Convert the range to a Table: Ctrl+T
- Single sort: City A to Z
- Multi-level sort: City, then OrderDate newest first (level order matters)
- Custom sort: Low, Medium, High

- Warning: click one cell inside the table; never sort a single column alone

Slide 24: Filtering

- Text filters: begins with, contains
- Number filters: greater than, top 10
- Date filters: this month, between
- Filter by colour to find highlighted cells
- Sheets: `=SORT()` and `=FILTER(A2:H, C2:C="Chennai")`

Slide 25: HANDS-ON 3A: Sort and Filter (8 min)

1. Show only Chennai orders, newest first
2. Show orders with Qty greater than 2
3. Copy the filtered results to a new sheet

Slide 26: Why Lookups?

- Orders has a product code but not the product name
- Products holds the name
- A lookup brings them together
- Four ingredients: what to find, where to look, what to return, exact match or not

Slide 27: VLOOKUP

```
=VLOOKUP(E2, Products!$A$2:$C$5, 2, FALSE)
```

- E2: lookup value
- Table range (lock with $); lookup value must be in the first column
- 2: column number to return
- FALSE: exact match (most common mistake: forgetting it)
- Cannot look left; breaks if columns are inserted

Slide 28: XLOOKUP (Preferred)

```
=XLOOKUP(E2, Products!$A$2:$A$5, Products!$B$2:$B$5, "Not found")
```

- Separate lookup and return columns
- Exact match by default

- Built-in not-found message
- Can look in any direction and return many columns
- P09 returns "Not found": the lookup caught a bad code

Slide 29: Lookup Errors

Error	Usual cause
#N/A	Value not found; trailing spaces; text vs number mismatch
#REF!	Column number too large or column deleted
#VALUE!	Wrong data type in the formula

Fix: TRIM the lookup column; make both sides the same data type

Slide 30: HANDS-ON 3B: Lookups (20 min)

1. Add **ProductName** and **Category** using XLOOKUP
2. Repeat ProductName with VLOOKUP; confirm both match
3. Add **Line_Total** = Qty x Price_Clean
4. Find the order with an unknown product code and write a fix recommendation
5. Challenge: return ProductName and Category in one XLOOKUP

Module 4: Power Query

Slide 31: What Is Power Query?

- Records your cleaning steps once
- Replays them on new data with one click: Refresh
- Think of it as a recipe you cook once and repeat forever
- Idea: Connect, Transform, Combine, Load
- Found under Data, Get Data (Excel only; also in Power BI)

Slide 32: The Editor

- Ribbon at the top

- Query list on the left
- Data preview in the middle
- **Applied Steps** on the right: your recipe, editable and reorderable
- Start: click inside the table, Data, From Table/Range

Slide 33: Data Sources

- Excel workbook, Text/CSV, Web
- Folder: combines many monthly files at once
- Rename each query clearly (Orders, Products)

Slide 34: Cleaning Steps

Task	Where
Remove spaces	Transform, Format, Trim
Fix capitalisation	Transform, Format, Capitalize Each Word
Fix types	Click the type icon on the column header
Replace values	Transform, Replace Values ("Rs." to nothing, "N/A" to null)
Remove duplicates	Home, Remove Rows, Remove Duplicates
Remove errors	Home, Remove Rows, Remove Errors
Split column	Transform, Split Column, By Delimiter
New column	Add Column, Custom Column: `[Qty] * [Price]`

Order matters: replace "Rs." first, then set the type to number

Slide 35: HANDS-ON 4A: Clean in Power Query (20 min)

1. Load Orders_Raw into Power Query
2. Trim and Capitalize Each Word on Customer and City
3. Replace "Rs." and "N/A"
4. Set data types (use English (India) locale for dates)
5. Remove duplicates on OrderID

6. Remove rows where Qty is less than 1
7. Rename the query **Orders**; keep the editor open

Slide 36: Merge Queries

- Merge adds columns by matching a key (like VLOOKUP, but stronger)
- Home, Merge Queries: ProductCode in Orders and Products

Join kind	Result
Left Outer	All Orders rows plus matches (best for lookups)
Inner	Only matching rows
Full Outer	Everything from both
Left Anti	Orders with no match (finds bad codes)

Slide 37: Append Queries

- Append stacks rows from tables with the same columns
- Example: January and February orders
- Home, Append Queries as New
- Column names must match exactly

Slide 38: Load and Refresh

- Home, Close and Load
- Add a new messy row to the raw sheet
- Right-click the output table, Refresh
- The new row is cleaned automatically
- Tip: make the source a Table (Ctrl+T) so it expands

Slide 39: HANDS-ON 4B: Merge, Append, Refresh (20 min)

1. Merge Orders with Products (Left Outer); expand ProductName and Category
2. Create **Bad_Codes** with a Left Anti merge
3. Create Orders_Feb (same eight columns, two rows) and append it
4. Close and Load; add a messy row to the raw sheet and Refresh

5. Challenge: open View, Advanced Editor and notice each step is a line of code

Check: Merge adds columns; Append adds rows

Module 5: AI-Assisted Data Preparation

Slide 40: Privacy First

- Never paste real or sensitive data into an AI tool
- No real names, emails, phone numbers, IDs or confidential client files
- Use sample or anonymised data only
- Keep 5 to 10 representative rows
- Rule: you are responsible for the result, not the AI

Slide 41: How to Anonymise

- Replace names with Customer A, Customer B
- Replace emails with user1@example.com
- Describe the column instead of pasting it
- Include a few messy rows so the AI sees the problems

Slide 42: A Good Prompt Has Four Parts

1. **Context:** Excel 365, Google Sheets or Power Query
2. **Data shape:** what the column contains
3. **Task:** what you want
4. **Output format:** formula, steps, explanation

- Weak: "Clean my data."
- Strong: "I use Excel 365. Column B has names with extra spaces and mixed capitalisation. Give one formula for C2 that returns Proper Case with single spaces, and explain each part."

Slide 43: Prompt Templates

1. Formula: "Column G has prices like 'Rs.450' and 'N/A'. Write an Excel formula that returns a number, blank for N/A."

2. Cleaning plan: "Here is an anonymised sample [paste]. List the problems and Power Query steps in order."
3. Explain: "Explain this formula line by line for a beginner."
4. Debug: "My XLOOKUP returns #N/A though the value exists. List causes and how to check each."
5. Validation: "Suggest validation rules for a mobile number column."

Slide 44: Verify Every Answer

- Test on three rows where you know the answer
- Test edge cases: blank, text, negative
- Compare row counts and totals before and after
- Understand every step; ask the AI to explain
- Check it works in your Excel version
- Never run code or macros you do not understand

Slide 45: HANDS-ON 5: AI Prompting (10 min)

1. Anonymise 6 rows of the practice data
2. Write a strong prompt for a formula that extracts the email domain
3. Paste the AI formula into Excel; test it on three rows
4. Write one sentence: was it correct, and what did you change?
5. Swap with a partner and critique each other's prompt

Check: What three things must you do before pasting data into AI?

Module 6: Capstone and Wrap-up

Slide 46: Capstone Challenge (20 min)

You get a fresh messy dataset (student marks list with duplicate roll numbers, mixed name cases, marks stored as text, missing marks, mixed dates of birth, department codes). In pairs:

1. Write an Issue Log
2. Clean with formulas or Power Query, with reasons
3. Add data validation to stop repeat problems

4. Add department names with XLOOKUP or a merge
5. Show a before and after count (rows, blanks, duplicates)
6. Use AI once with anonymised data and note how you verified it

Slide 47: Scoring Rubric (10 points)

Criterion	Points
Issue log completeness	2
Correct cleaning	3
Validation added	1
Lookup or merge correct	2
Verification and documentation	2

Slide 48: Quick Reference

Problem	Fix
Extra spaces	TRIM / Power Query Trim
Wrong case	PROPER, UPPER, LOWER
Number as text	VALUE, Text to Columns, set type
Mixed dates	Text to Columns, DATEVALUE, locale
Duplicates	COUNTIFS flag, Remove Duplicates, UNIQUE
Blanks	Go To Special, COUNTBLANK, Fill Down
Invalid entries	Data Validation
Combine by column	XLOOKUP, VLOOKUP, Merge
Combine by rows	Append
Repeat weekly	Power Query Refresh

Slide 49: Key Takeaways

- Keep raw data safe; clean a copy
- Decide rules first, then clean, then verify
- Prevent problems with validation
- Automate repeat work with Power Query
- AI: anonymise first, ask clearly, verify always

Slide 50: Homework

1. Clean a 50-row dataset of your own (attendance, lab inventory, scores); submit Issue Log, cleaned file and a one-page note on decisions
2. Build a Power Query combining two months of data; add a third month and Refresh
3. Write three AI prompts using anonymised data and submit screenshots of your verification

Slide 51: Exit Questions and Thank You

- One tool I will use tomorrow
- One thing that confused me
- One data problem from my own work I want to solve
- Thank you | Questions?